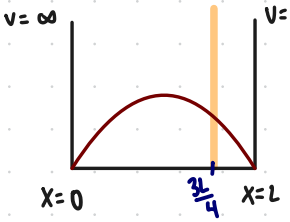
1. In class we computed the electron eigenstate wave function in a box with infinite barriers, with x ranging from 0 to L (in one dimension). For the lowest energy eigenstate, calculate the probability that if a measurement is done to find the electron, that the electron is found in the range 3 L /4 to L.

Infinite Squarewell



$\psi(x) = A \sin kx$

$\lambda = \frac{2L}{N}$

- $\psi(x) = A\sin\frac{2\pi x}{\left(\frac{2L}{N}\right)} = A\sin\left(\frac{N\pi x}{L}\right)$

· Find A by normalization

* total probability must equal 100% *

$$\boxed{1} = \int dx\,|\psi(x)|^2 = \int_0^L dx\, A^2 \sin^2\left(\frac{\pi x}{L}\right) dx$$

try: $\sin^2\theta = \frac{1-\cos 2\theta}{2}$: $\int ax\cos bx = \frac{1}{b}\sin bx$

$$\hookrightarrow A^2\left(\frac{L}{2} - \frac{L}{\pi}\cancelto{0}{\sin\frac{\pi x}{L}}\Big|_0^L\right) \Rightarrow \boxed{1} = A^2\frac{L}{2} \rightarrow A = \sqrt{2/L}$$

$$P = \int_{3L/4}^{L} A^2\sin^2\left(\frac{\pi x}{L}\right)dx < 1 : A^2\left[\frac{x}{2} - \frac{L}{4\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_{\frac{3L}{4}}^{L} \quad \text{where } A = \sqrt{2/L}$$

$$\hookrightarrow \frac{2}{L}\left[\frac{x}{2} - \frac{L}{4\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_{\frac{3L}{4}}^{L} < 1$$

$$P = \frac{2}{L}\left[\frac{x}{2} - \frac{L}{4\pi}\sin\left(\frac{2\pi x}{L}\right)\right]\Bigg|_{\frac{3L}{4}}^{L}$$

$$\hookrightarrow P = \frac{2}{L}\left[\frac{L}{2} - \left(\frac{3L}{8}\right) + \frac{L}{4\pi}\right] \Rightarrow \frac{2}{L}\left(\frac{L}{8} - \frac{L}{4\pi}\right) = \frac{1}{4} - \frac{1}{2\pi}$$

2. In class we found the normalization factor for an electron wave function for a box with infinite barriers with x ranging from 0 to L (in one dimension), in the lowest energy eigenstate. Calculate the normalization factor for the $N = 2$ eigenstate.

normalization - $\int_L dx\,|\psi|^2 = 1$

$\frac{1}{L}$

$\psi(x) = A \sin kx$

$\lambda = \frac{2L}{N}$

- $\psi(x) = A\sin\frac{2\pi x}{\left(\frac{2L}{N}\right)} = A\sin\left(\frac{N\pi x}{L}\right) = A\sin\left(\frac{2\pi x}{L}\right) \Rightarrow \int_0^L A^2\sin^2\left(\frac{2\pi x}{L}\right)dx \Rightarrow 1 : A^2\int\left[\frac{1}{2} - \frac{1}{2}\cos\left(\frac{4\pi x}{L}\right)\right]dx = 1$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\hookrightarrow \int ax\cos bx = \frac{1}{b}\sin bx \quad \hookrightarrow A^2\cdot\left[\frac{x}{2} - \frac{1}{2}\sin\left(\frac{4\pi x}{L}\right)\right]_0^L$

$$A^2\left[\frac{L}{2} - \cancelto{0}{\frac{1}{2}\sin(4\pi)}\right] = A^2\frac{L}{2} = 1,\ A = \sqrt{\frac{2}{L}}$$

3. Calculate the normalization factor for the lowest energy eigenstate of a well with finite barriers that we did in class, namely, x ranging from $-a/2$ to $a/2$, with potential energy $U = 0$ inside this range, and $U = U_0$ outside this range.

outside well:

$$\mathcal{E}\psi(x) = -\frac{\hbar}{2m}\frac{d^2}{dx}\psi(x) + U(x)\psi(x) \qquad \psi = \psi(x)e^{-i\omega t}$$

$$\hookrightarrow \mathcal{E}\cancel{\psi} = \frac{-\hbar^2 k^2 \cancel{\psi}}{2m} + U(x)\cancel{\psi}$$

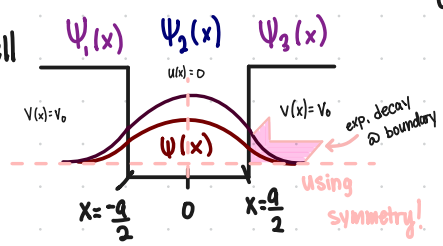
$$\psi(x) \propto e^{iKx} = e^{i(ik)x} = \underline{B}e^{-kx}$$

where:

$$K = \sqrt{\frac{2m(E-U)}{\hbar^2}}$$

$$\frac{d^2\psi}{dx^2} = K^2\psi$$

finite square well



@ lowest energy state: $U(x) = 0$ when $U(x) < \frac{a}{2}$

$U(x) = U_0$ when $U(x) > -\frac{a}{2}$

↳ special property:

(from symmetry) → $B = \pm A$, $b_3 = \pm a_1$

in symmetric case: $B = A$

outside the well:

$\psi_0(x) = Ae^{Kx} + Be^{Kx}$, for $x \to \infty$: $\psi \to \infty$: must set $A = 0$

↳ Be^{-Kx}, @ $x \geq \frac{a}{2}$: $\psi_0(\frac{a}{2}) = Be^{-\frac{ka}{2}}$

but, inside the well $x \leq \frac{a}{2}$: $\psi_i(\frac{a}{2}) = A\cos(kx)$

→ @ $x \equiv \frac{a}{2}$: $\psi_0(\frac{a}{2}) = \psi_i(\frac{a}{2}) = Be^{-\frac{ka}{2}} = A\cos(kx)$, so: $B = A\cos(kx)/e^{-ka/2} = A\cos(kx)\cdot e^{ka/2}$

so we plug back into: $\psi(x) = Be^{-Kx}$ and resolve $\psi(x) = A\cos(\frac{ka}{2})e^{\frac{ka}{2}}e^{-Kx} = A\cos(\frac{ka}{2})e^{-k(x-a/2)}$

@ boundary → $x = \frac{a}{2}$: $\psi(\frac{a}{2}) = A\cos(\frac{ka}{2})e^0 = A\cos(\underbrace{\tfrac{ka}{2}}_{\hookrightarrow x})$ w/ $a = \infty, \frac{a}{2}$ $b = \frac{a}{2}, 0$

normalization constant: since by symmetry: $0 \to \frac{a}{2} = \frac{-a}{2} \to 0$: resolving $-1 = 2\int_b^a |\psi|^2 dx + 2\int_b^a |\psi|^2 dx$

we multiply by 2

$$\Rightarrow 2\int_{\frac{a}{2}}^{\infty}|\psi|^2 dx + 2\int_0^{a/2}|\psi|^2 dx \Rightarrow \overset{A}{2\int_{a/2}^{\infty} A^2\cos^2(kx)\,dx} + \overset{B}{2\int_{\frac{a}{2}}^{0} A^2\cos^2\left(\frac{ka}{2}\right)e^{-k(x-\frac{a}{2})}} \quad \text{and} \quad \int_{-\infty}^{\infty}|\psi(x)|^2 dx = 1$$

A.)

$$\Rightarrow 2A^2\cos^2\left(\frac{ka}{2}\right)\int_{a/2}^{\infty} e^{-2k(x-a/2)}\,dx \Rightarrow \int_0^{\infty} e^{-2ku}\,du = \left[\frac{e^{-2ku}}{-2k}\right]_0^{\infty} = \frac{0-1}{-2k} = \frac{1}{2k} \Rightarrow A^2\cos^2\left(\frac{ka}{2}\right)\frac{1}{k}$$

$u = x - \frac{a}{2}$

@ $x = \frac{a}{2}$, $u = 0$; @ $x = \infty$, $u = \infty$ } $du = dx$

$$\int_0^{\frac{a}{2}} A^2\cos^2(kx)\,dx = A^2\int_0^{\frac{a}{2}}\frac{1-\cos(2kx)}{2}\,dx = \frac{A^2}{2}\int_0^{\frac{a}{2}} 1-\cos(2kx)\,dx = \frac{A^2}{2}\left[x + \frac{\sin(2kx)}{2k}\right]_0^{\frac{a}{2}}$$

$$* \cos^2(kx) = \frac{1-\cos(2kx)}{2}$$

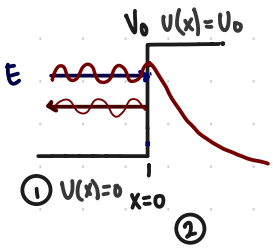
$$\hookrightarrow \frac{A^2}{2}\left[\frac{a}{2} + \frac{\sin(ka)-\sin(0)}{2k}\right] = \frac{A^2}{2}\left[\frac{a}{2} + \frac{\sin(ka)}{2k}\right]\cdot 2 = A^2\left(\frac{a}{2} + \frac{\sin(ka)}{2k}\right)$$

together:

$$1 = A^2\left(\frac{a}{2} + \frac{\sin(ka)}{2k}\right) + \left[A^2\cos\left(\frac{ka}{2}\right)\right]\frac{1}{k} = A^2\left(\frac{a}{2} + \frac{\sin(ka)}{2k} + \frac{\cos^2\left(\frac{ka}{2}\right)}{k}\right)$$

$$\text{so } A = \left(\frac{a}{2} + \frac{\sin(ka)}{2k} + \frac{\cos^2\left(\frac{ka}{2}\right)}{k}\right)^{-\frac{1}{2}}$$

4. In class we looked at the case of "scattering" from a finite barrier, with a wave with wavenumber k coming from the left in a region with $U = 0$, hitting a region starting at $x = 0$ with $U = U_0$, when the kinetic energy E is greater than U_0. Calculate what happens when the wave coming from the left has E <u>lower</u> than U_0. In this case there will not be a propagating wave going off to the right, there will be an evanescent wave on that side, and a reflected wave going back to the left.



$$\mathcal{E}\psi(x) = -\frac{\hbar}{2m}\frac{d^2}{dx}\psi(x) + U(x)\psi(x)$$

superposition principle:

region 1:

$$\psi_1 = Ae^{ikx} + Be^{-ikx}$$

$$\psi_2 = \frac{-\hbar^2}{2m}\psi_2'' + U_0 E\psi_2$$

$$= \frac{-\hbar^2}{2m}\psi_2'' \Rightarrow \psi_2'' - k^2\psi_2 = 0 \Rightarrow Ae^{-kx} + \cancel{Be^{kx}} = Ae^{-kx}$$

$$\psi_1(0) = \psi_2(0)$$

$$k' = \frac{\sqrt{2m(E-V_0)}}{\hbar} \qquad E = \frac{\hbar^2 k^2}{2m}$$

$$\lambda = \frac{2\pi}{k}, \quad k \text{ is input}$$

Boundary @ $x=0$

$$\psi_1 = \cancel{e^{ik(0)}}^{\,1} + \cancel{re^{-ik(0)}}^{\,1} = te^{ik'(0)}$$

$$\Rightarrow \boxed{1+r=t}$$

$$ik\cancel{e^{ik(0)}}^{\,1} - ikr\cancel{e^{-ik(0)}}^{\,1} = ik't\cancel{e^{ik(0)}}^{\,1}$$

$$\hookrightarrow ik(1-r) = ik't$$

$$\boxed{k(1-r) = k't}$$

$$r = t-1$$

$$k(1-(t-1)) = k't$$

$$k(2-t) - k't = 0$$

$$2k - (k+k')t = 0$$

$$\rightarrow \boxed{t = \frac{2k}{k+k'}}$$

$$\hookrightarrow r = t-1 = \frac{2k' - (k+k')}{k+k'}$$

$$\hookrightarrow \boxed{r = \frac{k-k'}{k+k'}} \text{ aka } \frac{ik+k}{ik-k}$$

@ $x=0$: $1+r = A$:

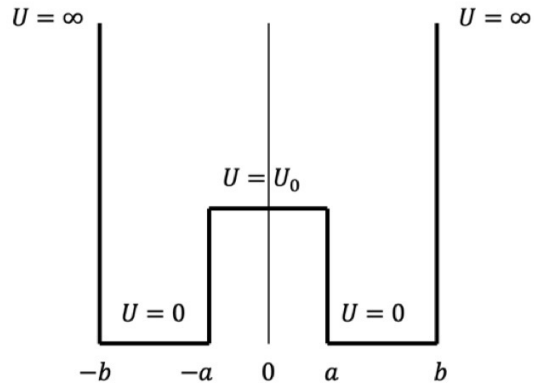
1st derivative: $ik(1-r) = kA$ and w/ $A = 1+r$: $ik(1-r) = k(1-r)$

$$\hookrightarrow = ik(1-r) = -Ak$$

Reflection Coefficient $R = |r|^2$

$$\hookrightarrow \frac{|k+ik|^2}{|-k+ik|^2} = \frac{k^2+k^2}{k^2+k^2} = 1$$

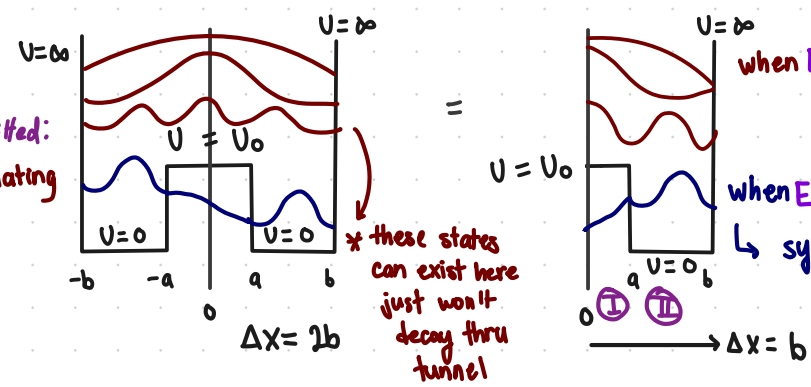
5. Write down the equations for finding the eigenstates of an electron wave confined in the potential energy profile shown below, but <u>do not solve them</u>. (Unless you feel so inclined for your own interest, and want to use a numerical program.)



How might the symmetry of this system be used to make this solution easier?

because $U(x)$ is symmetric about $x=0$ we get def. parity where $\psi(x) = \psi(-x)$ about $x=0$ for even and $\psi(x) = \psi(x)$ about $x=0$ for odd

eigenstates permitted:
when $E > U_0 \rightarrow$ oscillating
when $E < U_0$
↳ tunneling



* these states can exist here just won't decay thru tunnel

Region II ($\Delta x = a$)
Region I ($\Delta x = b-a$)

↳ symmetry allows us to solve for $[0, b]$ and reflect for $[0, -b]$

equations can be reduced to 2 regimes: ① determined by $\Delta x = b$ and ② by a (where $E < U_0$) and b

Region Ⅰ: sinosuiodal if $E > U_0$ (oscillating $E > U_0$)
exp. if $E < U_0$ (tunneling / evanescent $E < U_0$)

* because $U = \infty$ @ b: $\psi(b) = 0$

exp. decay @ $0 \leq x \leq a$
scillating @ $a \leq x b$)

Region Ⅱ sinosuidal

in the well $= k_1 = \sqrt{\frac{2mE}{\hbar^2}}$ (in region I)

@ $E < U_0$: $k_2 = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}$

and

@ $E > U_0$: $q = \sqrt{\frac{2m(E - U_0)}{\hbar^2}}$

@ $U = 0$: $\psi + k_1^2 \psi = 0$

↳ need: $\psi(b) = 0$: $\psi_{II} = A \sin(k(b - x))$